SHRIVIDYA HEGDE

Forward Deployed AI Engineer | Production LLM & RAG Systems | Apache Airflow Champion, #1 Globally Q3 2026

Atlanta, GA | shrividyahegde222@gmail.com | linkedin.com/in/shrividya-hegde-shri-91562365 | github.com/Shrividya | shrividya.github.io

Professional Summary

Engineer who takes LLM and RAG systems from prototype to production inside complex, multi-team environments, then stays to make them reliable. Five years building orchestration, ETL/ELT and production LLM systems across AWS, GCP and Azure, on top of seven years in test automation and quality engineering for banking and enterprise clients. At Astronomer, the company behind Apache Airflow, I grew from SDET into the sole engineer owning quality and release reliability for Astro Observe and end-to-end Airflow 3 regression testing. I work directly with product, platform and client stakeholders, scope ambiguous problems quickly, ship working integrations, and explain the tradeoffs in plain language to technical and non-technical audiences.

Key Achievements

- Built and deployed an LLM-powered RAG system (LangChain, ChromaDB) that summarizes test logs for release gating, flags defects and routes findings to developers over Slack, improving issue resolution speed approximately 40%.
- Sole engineer owning quality and release reliability for Astro Observe, Astronomer's managed Airflow observability product, and end-to-end owner of Airflow 3 regression testing.
- Increased pipeline throughput approximately 30%, cut data delivery time by over 20%, and reduced cloud infrastructure cost approximately 60% across multi-cloud orchestration workloads.
- Ranked #1 Apache Airflow Champion globally in Q3 2026 and Top 5 in Q2 2026; 7 merged pull requests to Apache Airflow core.
- Speaker at Apache Airflow Summit 2026 and the Apache Airflow NYC Meetup ("Airflow + AI: Orchestrating the Intelligent Stack"); mentor with 20+ bookings in a month and nine 5-star testimonials.

Technical Skills

LLM & AI Systems: LLM integration, Retrieval Augmented Generation (RAG), LangChain, OpenAI API, prompt engineering, vector databases (ChromaDB, FAISS), semantic search, NLP (NLTK), MLOps, generative AI, agent deployment and cost control

Orchestration & Data Engineering: Apache Airflow (3.x and 2.x), Astronomer, DAG authoring, dynamic task mapping, ETL/ELT, data pipelines, dbt, Apache Spark and PySpark, Snowflake, PostgreSQL, data modeling and warehousing, batch processing

Cloud & Infrastructure: AWS, Google Cloud Platform, Microsoft Azure, Kubernetes, Docker, Terraform and infrastructure as code, boto3, Azure SDK, Google Cloud SDK, Astro SDK

Reliability, Quality & Observability: Production monitoring, release gating, SLA management, data quality and reliability engineering, data lineage and OpenLineage, data governance, performance optimization, distributed systems, automated regression testing

Integration & Delivery: REST APIs, FastAPI, Streamlit, Slack integrations, GitHub Actions, CircleCI, GitLab CI, Azure DevOps, Git

Programming: Python, SQL, PySpark, Bash, Java

Professional Experience

Senior AI Data Engineer (Freelance) | Self-employed | Remote, United States | September 2026 to Present

- Deliver Apache Airflow consulting and AI data engineering projects for clients, covering pipeline design and review, RAG systems and orchestration of AI workloads.

Open Source Contributor, Apache Airflow | The Apache Software Foundation | Remote | March 2026 to Present

- Shipped 7 merged pull requests to Apache Airflow core across dynamic task mapping, scheduler improvements, bug fixes and documentation.
- Named Apache Airflow Champion (January 2026) for contributions, pull request reviews and issue triage; ranked Top 5 globally in Q2 2026 and #1 globally in Q3 2026.
- Take part in Airflow 3 feature development and governance discussions; review pull requests for Apache Spark.

Senior Software Engineer, Astro Observe (Data Platform) | Astronomer | Bengaluru, India (Remote) and Parsippany, NJ | March 2022 to November 2025

- Joined as an SDET and moved into data platform and AI engineering; finished as the sole engineer owning quality and release reliability for Astro Observe and end-to-end owner of Airflow 3 regression testing, working with the Astro Build and Run teams on Airflow 3 adoption.
- Built and deployed an LLM-powered RAG system (LangChain, ChromaDB) that summarized test logs for release gating, flagged defects and routed findings to the right developers over Slack, improving issue resolution speed approximately 40%.
- Architected cloud-native data pipeline frameworks for Astro in Python with Airflow DAGs and cloud SDKs (boto3, azure-sdk, gcp-sdk, astro-sdk), increasing pipeline throughput approximately 30%.
- Designed and tuned ETL/ELT orchestration on Apache Airflow and Kubernetes across GCP, Azure and AWS, cutting data delivery time by over 20%.
- Wrote the Airflow DAGs that drive release gating and sign-off for the Observability team, giving every team visibility into release status on each commit.
- Rebuilt the QA automation framework into the Astro release pipeline and embedded data quality and observability checks in CI/CD (CircleCI, GitHub Actions) to protect pipeline reliability and data integrity at enterprise scale.
- Automated cloud resource lifecycle management with GCP and AWS CLI tooling, reducing infrastructure cost approximately 60%.

Selected Projects

- **Resume Analyser RAG** (LangChain, FAISS, OpenAI API, Python): multi-document retrieval and semantic search pipeline that answers natural-language questions over uploaded resumes. Code
- **Article Insights NLP Pipeline** (Apache Airflow 3, Python, NLTK): production-grade pipeline running parallel DAG tasks for keyword extraction, summarization, sentiment analysis, entity recognition and readability scoring, with structured JSON output. Code
- **OpenAI Chatbot** (OpenAI API, Python): chatbot demonstrating system message design, context management and prompt engineering patterns. Code
- **Real-Time Stock Sentiment Analyzer** (Python, Streamlit, Docker, GitHub Actions): containerized market and news sentiment dashboard with CI/CD deployment. Code

Earlier Experience: Test Automation and Quality Engineering

Senior Test Automation Engineer, Core Payments Data | Commonwealth Bank of Australia | Bengaluru, India | May 2021 to February 2022

- Led automation strategy for Core Payments data systems across a microservices architecture, reaching 95% automation coverage and cutting testing costs 15%.
- Designed reusable data validation frameworks for pipeline verification across payment services; awarded Best Partner for automation framework delivery.

Consultant, Automation Specialist | KPMG India | Bengaluru, India | May 2020 to May 2021

- Built automation frameworks adopted by 4+ delivery teams, reducing manual QA cycles 25% and lifting framework adoption 30%; was the SME and day-to-day stakeholder contact for test automation on the Shell India client account.
- Built Azure infrastructure automation with Terraform for Shell India; configured Azure Data Factory pipelines and data anonymization workflows feeding Power BI dashboards for the Center of Excellence data team.
- Delivered a SAP S/4 HANA automation proof of concept using BrowserStack and Appium, with data readiness and governance recommendations; published reusable utilities as NuGet and JAR packages with CI/CD on Azure DevOps.

IT Analyst / Software Engineer | Tata Consultancy Services | Bengaluru, India and Sydney, Australia | June 2014 to April 2020

- Built unified automation frameworks for CI/CD across Banking and Finance data platforms, cutting manual testing effort 50% and raising deployment frequency 35%.
- Developed Java tooling for SQL and PL/SQL data operations, reducing manual database intervention 60%. Received 5 internal awards including the Service Commitment Award and Star of the Month.

Speaking, Mentoring and Community

- **Director, Women in Big Data, Atlanta Chapter** (October 2026 to Present): lead the Atlanta chapter's programming, events and community growth for women in data and AI.
- **Speaker:** Airflow Online Summit Reconnect, November 5, 2026 (upcoming, "Self-Healing Airflow Pipelines"); Apache Airflow Summit 2026; Apache Airflow NYC Meetup, June 2026 ("Airflow + AI: Orchestrating the Intelligent Stack"); technical

session for the dbt community; hackathon judge for Astronomer and Women in Data. Talks on Sessionize

- **Women in Data, New Jersey Chapter Lead** (March 2025 to Present): run technical bootcamps on SQL, Python, PySpark and applied machine learning, and mentor engineers one to one.
- **Expert Mentor, Data & AI, Ediso** (2026 to Present) and **mentor on Topmate** (2026 to Present): advise engineers on data engineering, orchestration, applied AI and career transitions; 20+ bookings in a month and nine 5-star testimonials.
- **Technical writing:** articles on Medium and Substack covering RAG architecture, LLM agent deployment, Kubernetes cost control and Airflow cost optimization; expert content writer for Data Engineering Byte. Podcast guest on The DataFlow Cast; nominated for the Women Tech Network Global Awards. Writing

Certifications

- **AI and cloud:** AI Engineer for Developers Associate (DataCamp), July 2026; AWS Certified Cloud Practitioner, June 2025; Generative AI Fundamentals (Databricks Academy), March 2025; AI Fundamentals (DataCamp), 2025
- **Data engineering:** Astronomer Champions Program, January 2026; DAG Authoring and Fundamentals for Airflow 3, June 2025, and Airflow 2, March to May 2025; Data Engineer Associate (DataCamp), April 2026; Python Data Associate (DataCamp), May 2026
- **Additional:** Python for Data Science (DataCamp); Java Certified (HackerRank); Certified Selenium Professional (Vskills)

Education

Bachelor of Engineering, Computer Science and Engineering, Malnad College of Engineering, 2010 to 2014, CGPA 9.06 / 10. WES credential evaluation completed (Canada).